

Homework 2 - Guaranteed and Dynamic VWAP Strategies

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Problem 1

Overview

This project studies guaranteed VWAP (Volume Weighted Average Price) execution strategies using historical quote data from the TAQ dataset. We simulate both static and dynamic strategies to fulfill client orders across a 13-bucket trading day while minimizing market impact and execution costs.

Problem Setup

We consider a broker who receives a client order to execute X shares of a given stock (e.g., GE or AAPL) over a trading day. The objective is to execute the trade in a way that guarantees a price no worse than the VWAP benchmark, possibly with a premium to account for slippage and impact risk.

The key challenge lies in allocating trades across the day in response to observed market volume while anticipating future liquidity.

Static Strategy

We first define a baseline static strategy where the execution profile is proportional to the average historical volume pattern:

$$x_t = \frac{V_t}{\sum_{s=1}^{13} V_s} \cdot X,$$

where V_t is the average volume in bucket t and X is the total number of shares to execute.

This strategy assumes the trader has no information about intraday variations and executes in proportion to long-term expectations.

Dynamic Strategy

To improve upon the static strategy, we build a dynamic simulator that adapts execution speed based on deviations between observed and forecasted volume.

Forecast Model

We inject noise into the static profile to simulate forecast uncertainty. The volume forecast at each step t is:

$$\hat{v}_t = \frac{v_t + \varepsilon_t}{\sum_{s=t}^{13} (v_s + \varepsilon_s)},$$

where $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$.

Execution Algorithm

At each bucket t :

- If $t = 0$, the simulator trades according to the static profile.
- Otherwise, it calculates the deviation between cumulative observed and forecasted volume:

$$\delta_t = \frac{\sum_{s=0}^{t-1} v_s^{\text{obs}} - \sum_{s=0}^{t-1} \hat{v}_s}{\sum_{s=0}^{t-1} \hat{v}_s + \epsilon}.$$

- This deviation adjusts the urgency parameter in a sinh-based trajectory function:

$$w_t \propto \sinh(\gamma(T - t + 1)),$$

where γ is dynamically adjusted by urgency based on δ_t .

We ensure the remaining shares are fully executed by the end of the day and enforce minimum trading floors in each bucket.

Execution Cost Modeling

The execution cost incorporates temporary market impact via a linear impact model:

$$\text{ExecPrice}_t = \text{MidQuote}_t + \eta \cdot \text{ParticipationRate}_t,$$

with η calibrated from TAQ data. The total execution cost is computed and compared against the VWAP benchmark of the day.

Empirical Results

We evaluate both strategies on historical TAQ data using a training-testing split.

Static Model Evaluation

We report model fit on volume patterns:

- Mean Absolute Error (MAE): 0.0187
- Root Mean Squared Error (RMSE): 0.0294
- R^2 : -0.06 (limited predictive power)

Execution Performance

On multiple test days, the dynamic strategy adapts more aggressively in the presence of unexpected liquidity (e.g., increased volume in early buckets), reallocating execution accordingly.

The guaranteed VWAP quote computed using dynamic execution consistently matches or outperforms static strategy pricing, with controlled slippage and minimized terminal trade imbalances.

Conclusion

This project demonstrates a full pipeline from raw TAQ quote data to dynamic intraday execution simulation. By incorporating volume forecast deviations and adjusting execution urgency dynamically, we achieve a more realistic and adaptive trading strategy that improves upon naive static execution.

Problem 2

The formulas given:

$$\begin{aligned} dQ_t &= -\nu_t dt, & Q_0 &= N, \\ dS_t^\nu &= -b\nu_t dt + \sigma dW_t, \\ \hat{S}_t &= S_t^\nu - k\nu_t, \\ dX_t &= \hat{S}_t \nu_t dt, \\ V_t &= \int_0^t \mu_s ds, \\ \delta_t &= (1 - \rho)(Q_0 - Q_t) - \rho V_t. \end{aligned}$$

For any admissible trading rate ν , define strategy performance evaluation as

$$H^\nu(t, x, S, V, q) = \mathbb{E}_{t,x,S,V,q} \left[X_T + Q_T(S_T^\nu - \alpha Q_T) - \varphi \int_t^T \delta_s^2 ds \right], \quad H(t, \cdot) = \sup_\nu H^\nu.$$

(a) Derivation of the HJB Equation

Define the value function:

$$H(t, x, S, V, q) = \sup_\nu \mathbb{E}_{t,x,S,V,q} \left[X_T + Q_T(S_T^\nu - \alpha Q_T) - \varphi \int_t^T \delta_s^2 ds \right], \quad H(T, x, S, V, q) = x + q(s - \alpha q)$$

where the control is the trading rate ν .

We seek the HJB equation satisfied by H . By Itô's Lemma,

$$dH = \partial_t H dt + \partial_x H dX_t + \partial_S H dS_t + \partial_V H dV_t + \partial_q H dQ_t + \frac{1}{2} \partial_{SS} H (dS_t)^2.$$

Recall the state dynamics:

$$\begin{aligned} dX_t &= (S_t - k\nu_t)\nu_t dt, \\ dS_t &= -b\nu_t dt + \sigma dW_t, \\ dV_t &= \mu_t dt, \\ dQ_t &= -\nu_t dt. \end{aligned}$$

The expectation of the increment for a control ν_t , is:

$$\mathbb{E}[dH] = \partial_t H dt + \partial_x H dX_t + \partial_S H \mathbb{E}[dS_t] + \partial_V H dV_t + \partial_q H dQ_t + \frac{1}{2} \partial_{SS} H \mathbb{E}[(dS_t)^2].$$

as only dS_t has stochastic part.

Plug in the increments:

$$\begin{aligned}\mathbb{E}[dS_t] &= -b\nu_t dt, \\ \mathbb{E}[(dS_t)^2] &= \sigma^2 dt, \\ dX_t &= (S_t - k\nu_t)\nu_t dt, \\ dV_t &= \mu_t dt, \\ dQ_t &= -\nu_t dt.\end{aligned}$$

Thus,

$$\mathbb{E}[dH] = \left[\partial_t H + \partial_x H (S_t - k\nu_t)\nu_t - b\nu_t \partial_S H + \mu_t \partial_V H - \nu_t \partial_q H + \frac{1}{2}\sigma^2 \partial_{SS} H \right] dt.$$

The running cost is $-\varphi\delta_t^2$ per unit time based on value function, so the dynamic programming principle says:

$$0 = \sup_{\nu} \left\{ -\varphi\delta_t^2 + \partial_t H + \partial_x H (S_t - k\nu_t)\nu_t - b\nu_t \partial_S H + \mu_t \partial_V H - \nu_t \partial_q H + \frac{1}{2}\sigma^2 \partial_{SS} H \right\}$$

But since H is linear in x at the end time (recall terminal condition $H(T, x, S, V, q) = x + q(s - \alpha q)$) and the time increment per unit in X_t is independent of x , we know $\partial_x H = 1$ at all times. Thus, substitute $\partial_x H = 1$:

$$0 = -\varphi\delta_t^2 + \partial_t H + \mu_t \partial_V H + \frac{1}{2}\sigma^2 \partial_{SS} H + \sup_{\nu} \left\{ (S_t - k\nu_t)\nu_t - b\nu_t \partial_S H - \nu_t \partial_q H \right\}$$

or, after tidying up notation and substituting in δ 's expression,

$$0 = \partial_t H - \varphi[(1 - \rho)(Q_0 - q) - \rho V]^2 + \mu \partial_V H + \frac{1}{2}\sigma^2 \partial_{SS} H + \sup_{\nu} \left\{ (S - k\nu)\nu - b\nu \partial_S H - \nu \partial_q H \right\}$$

This is the desired HJB equation.

(b) Show that the value function takes the quadratic form

Assume a quadratic ansatz for the value function:

$$H(t, x, S, \mu, V, q) = x + qS + h_0(t, V) + h_1(t, V)q + h_2(t)q^2$$

where $h_0(t, V)$, $h_1(t, V)$, and $h_2(t)$ are functions to be determined.

Plugging derivatives into the HJB and solving for optimal control

Recall the quadratic ansatz:

$$H(t, x, S, \mu, V, q) = x + qS + h_0(t, V) + h_1(t, V)q + h_2(t)q^2$$

with

$$\begin{aligned}\partial_t H &= h_{0,t} + h_{1,t}q + h_2'(t)q^2 \\ \partial_S H &= q \\ \partial_q H &= S + h_1 + 2h_2q \\ \partial_V H &= h_{0,V} + h_{1,V}q \\ \partial_{SS} H &= 0\end{aligned}$$

The HJB equation is:

$$0 = \partial_t H - \varphi [(1 - \rho)(Q_0 - q) - \rho V]^2 + \mu \partial_V H + \sup_{\nu} \{-b\nu \partial_S H + (S - k\nu)\nu - \nu \partial_q H\}$$

substitute the derivatives:

$$0 = [h_{0,t} + h_{1,t}q + h'_2(t)q^2] - \varphi [(1 - \rho)(Q_0 - q) - \rho V]^2 + \mu[h_{0,V} + h_{1,V}q] + \sup_{\nu} \{-b\nu q + (S - k\nu)\nu - \nu(S + h_1 + 2h_2q)\}$$

Expand the supremum:

$$\begin{aligned} & -b\nu q + (S - k\nu)\nu - \nu(S + h_1 + 2h_2q) \\ &= -b\nu q + S\nu - k\nu^2 - S\nu - h_1\nu - 2h_2q\nu \\ &= -k\nu^2 - (bq + h_1 + 2h_2q)\nu \end{aligned}$$

So,

$$\sup_{\nu} \{-k\nu^2 - (bq + h_1 + 2h_2q)\nu\}$$

Set the derivative w.r.t. ν to zero:

$$\begin{aligned} \frac{\partial}{\partial \nu} [-k\nu^2 - (bq + h_1 + 2h_2q)\nu] &= -2k\nu - (bq + h_1 + 2h_2q) = 0 \\ \implies \nu^* &= -\frac{bq + h_1 + 2h_2q}{2k} \end{aligned}$$

Plug this back into the quadratic to get the supremum:

$$\begin{aligned} \sup_{\nu}(\dots) &= -k(\nu^*)^2 - (bq + h_1 + 2h_2q)\nu^* \\ &= -k \left(\frac{-(bq + h_1 + 2h_2q)}{2k} \right)^2 - (bq + h_1 + 2h_2q) \left(\frac{-(bq + h_1 + 2h_2q)}{2k} \right) \\ &= -k \frac{(bq + h_1 + 2h_2q)^2}{4k^2} + \frac{(bq + h_1 + 2h_2q)^2}{2k} \\ &= -\frac{(bq + h_1 + 2h_2q)^2}{4k} + \frac{2(bq + h_1 + 2h_2q)^2}{4k} \\ &= \frac{(bq + h_1 + 2h_2q)^2}{4k} \end{aligned}$$

Final form of the HJB with all terms plugged in:

$$0 = [h_{0,t} + h_{1,t}q + h'_2(t)q^2] - \varphi [(1 - \rho)(Q_0 - q) - \rho V]^2 + \mu[h_{0,V} + h_{1,V}q] + \frac{(bq + h_1 + 2h_2q)^2}{4k}$$

This expression can now be grouped by powers of q to solve for h_2 , h_1 , and h_0 .

Expand all terms and collect powers of q

For q^2 : Riccati ODE for h_2

The q^2 -term in the HJB yields the Riccati equation:

$$h_2'(t) + \frac{1}{k}h_2^2 + \frac{b}{k}h_2 + \frac{b^2}{4k} - \varphi(1-\rho)^2 = 0, \quad h_2(T) = -\frac{1}{2}\alpha$$

Define the new variable

$$y(t) := h_2(t) + \frac{b}{2}$$

so that

$$h_2'(t) = y'(t), \quad h_2^2 = \left(y - \frac{b}{2}\right)^2 = y^2 - by + \frac{b^2}{4}$$

Substitute into the ODE:

$$y'(t) + \frac{1}{k} \left[y^2 - by + \frac{b^2}{4} \right] + \frac{b}{k} \left(y - \frac{b}{2} \right) + \frac{b^2}{4k} - \varphi(1-\rho)^2 = 0$$

$$y'(t) + \frac{1}{k}y^2 - \frac{b}{k}y + \frac{b^2}{4k} + \frac{b}{k}y - \frac{b^2}{2k} + \frac{b^2}{4k} - \varphi(1-\rho)^2 = 0$$

Notice $-\frac{b}{k}y + \frac{b}{k}y$ cancels, and combine constants:

$$y'(t) + \frac{1}{k}y^2 + \left(\frac{b^2}{4k} + \frac{b^2}{4k} - \frac{b^2}{2k} \right) - \varphi(1-\rho)^2 = 0$$

$$y'(t) + \frac{1}{k}y^2 - \varphi(1-\rho)^2 = 0$$

This is a standard Riccati equation. The terminal condition is

$$y(T) = h_2(T) + \frac{b}{2} = -\frac{\alpha}{2} + \frac{b}{2}$$

Let $\xi := \sqrt{\frac{\varphi(1-\rho)^2}{k}}$. The Riccati equation is:

$$y'(t) + \frac{1}{k}y^2 - k\xi^2 = 0$$

Separate variables and integrate:

$$\frac{dy}{k\xi^2 - \frac{1}{k}y^2} = dt$$

$$\frac{dy}{k^2\xi^2 - y^2} = dt$$

Let $a := k\xi$, then $k^2\xi^2 = a^2$:

$$\int \frac{dy}{a^2 - y^2} = \int dt$$

The general solution is:

$$\frac{1}{2a} \ln \left| \frac{a+y}{a-y} \right| = T - t + C$$

Solve for y :

$$\frac{a+y}{a-y} = \gamma e^{2a(T-t)}$$

for some constant γ .

Solving for $y(t)$:

$$\begin{aligned} a+y(t) &= \gamma e^{2a(T-t)}(a-y(t)) \\ a+y(t) &= \gamma a e^{2a(T-t)} - \gamma y(t) e^{2a(T-t)} \\ y(t) + \gamma y(t) e^{2a(T-t)} &= \gamma a e^{2a(T-t)} - a \\ y(t)[1 + \gamma e^{2a(T-t)}] &= \gamma a e^{2a(T-t)} - a \\ y(t) &= a \frac{\gamma e^{2a(T-t)} - 1}{\gamma e^{2a(T-t)} + 1} \end{aligned}$$

Return to $h_2(t) = y(t) - \frac{b}{2}$ and $a = k\xi = \sqrt{k\varphi(1-\rho)^2}$:

$$h_2(t) = \sqrt{k\varphi(1-\rho)^2} \frac{\gamma e^{2\xi k(T-t)} - 1}{\gamma e^{2\xi k(T-t)} + 1} - \frac{b}{2}$$

In equivalent hyperbolic form:

$$h_2(t) = -\sqrt{k\varphi}(1-\rho) \cdot \frac{\gamma e^{\xi(T-t)} + e^{-\xi(T-t)}}{\gamma e^{\xi(T-t)} - e^{-\xi(T-t)}} - \frac{1}{2}b$$

where

$$\begin{aligned} \xi &= \sqrt{\frac{\varphi(1-\rho)^2}{k}}, \quad \gamma = \frac{\alpha - \frac{1}{2}b + \sqrt{k\varphi}(1-\rho)}{\alpha - \frac{1}{2}b - \sqrt{k\varphi}(1-\rho)} \\ h_2(t) &= -\sqrt{k\varphi}(1-\rho) \cdot \frac{\gamma e^{\xi(T-t)} + e^{-\xi(T-t)}}{\gamma e^{\xi(T-t)} - e^{-\xi(T-t)}} - \frac{1}{2}b \end{aligned}$$

with terminal condition $h_2(T) = -\frac{1}{2}\alpha$.

For q : Linear PDE for h_1

The q -coefficient yields a linear first-order PDE:

$$h_{1,t}(t, V) + \mu h_{1,V}(t, V) + \frac{b + 2h_2(t)}{2k} h_1(t, V) + 2\varphi((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u]) = 0, \quad h_1(T, V) = 0$$

Method of Characteristics:

Let V_u be the solution to $dV_u = \mu du$, $V_t = V$, so $V_u = V + \mu(u-t)$ if μ is deterministic.

Let $\tau = u$ along the characteristic. Then, the system of ODEs is:

$$\frac{d}{du} h_1(u, V_u) + \frac{b + 2h_2(u)}{2k} h_1(u, V_u) = -2\varphi((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u])$$

with terminal condition $h_1(T, V_T) = 0$.

This is a linear nonhomogeneous ODE in h_1 along each characteristic. The integrating factor is

$$\exp\left(\int_t^u \frac{b + 2h_2(s)}{2k} ds\right)$$

Define

$$\ell(u, t) := \exp\left(-\int_t^u \frac{b + 2h_2(s)}{2k} ds\right)$$

By integrating backwards, we can write the solution as:

$$\begin{aligned} h_1(t, V) &= \int_t^T \ell(u, t) \left[-2\varphi((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u]) \right] du \\ &= 2\varphi \int_t^T \ell(u, t) ((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u]) du \end{aligned}$$

where $\ell(u, t)$ can be computed explicitly for the Riccati solution h_2 , and $\mathbb{E}_{t,V}[V_u]$ is the expectation of V_u given $V_t = V$.

$$h_1(t, V) = 2\varphi \int_t^T \ell(u, t) ((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u]) du$$

where

$$\ell(u, t) := \exp \left(- \int_t^u \frac{b + 2h_2(s)}{2k} ds \right)$$

Constant term: Linear PDE for h_0

The constant term yields

$$h_{0,t}(t, V) + \mu h_{0,V}(t, V) + \frac{1}{4k} h_1(t, V)^2 - \varphi((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u])^2 = 0, \quad h_0(T, V) = 0$$

This is a linear first-order PDE that can be solved by the method of characteristics.

Along characteristics: $V_u = V + \mu(u - t)$, $u \geq t$.

Write the ODE along a characteristic:

$$\frac{d}{du} h_0(u, V_u) = -\frac{1}{4k} h_1(u, V_u)^2 + \varphi((1-\rho)Q_0 - \rho V_u)^2$$

with $h_0(T, V_T) = 0$.

Integrate backward from T to t :

$$h_0(t, V) = \int_t^T \left[\frac{1}{4k} h_1(u, V_u)^2 - \varphi((1-\rho)Q_0 - \rho V_u)^2 \right] du$$

If V is stochastic, take the conditional expectation over the path V_u :

$$h_0(t, V) = \int_t^T \mathbb{E}_{t,V} \left[\frac{1}{4k} h_1(u, V_u)^2 - \varphi((1-\rho)Q_0 - \rho V_u)^2 \right] du$$

where $\mathbb{E}_{t,V}[\cdot]$ denotes expectation given $V_t = V$.

(c) Closed-form Optimal Trading Speed

Recall,

$$\nu_t^* = -\frac{bq + h_1 + 2h_2q}{2k}$$

where

$$h_1(t, V) = 2\varphi \int_t^T \ell(u, t) ((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u]) du$$

and

$$h_2(t) = -\sqrt{k\varphi}(1-\rho) \cdot \frac{\gamma e^{\xi(T-t)} + e^{-\xi(T-t)}}{\gamma e^{\xi(T-t)} - e^{-\xi(T-t)}} - \frac{1}{2}b$$

Plug in h_1 and h_2 :

$$\begin{aligned} \nu_t^* &= -\frac{bq + h_1 + 2h_2q}{2k} \\ &= -\frac{bq}{2k} - \frac{h_1}{2k} - \frac{h_2q}{k} \\ &= -\frac{bq}{2k} - \frac{1}{2k} \cdot 2\varphi \int_t^T \ell(u, t) ((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u]) du \\ &\quad - \frac{1}{k} \left[-\sqrt{k\varphi}(1-\rho) \frac{\gamma e^{\xi(T-t)} + e^{-\xi(T-t)}}{\gamma e^{\xi(T-t)} - e^{-\xi(T-t)}} - \frac{1}{2}b \right] q \end{aligned}$$

Simplifying:

$$\begin{aligned} \nu_t^* &= -\frac{bq}{2k} - \frac{\varphi}{k} \int_t^T \ell(u, t) ((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u]) du \\ &\quad + \frac{\sqrt{k\varphi}}{k} (1-\rho) \frac{\gamma e^{\xi(T-t)} + e^{-\xi(T-t)}}{\gamma e^{\xi(T-t)} - e^{-\xi(T-t)}} q + \frac{1}{2k} bq \end{aligned}$$

Notice that $-\frac{bQ_t}{2k} + \frac{1}{2k}bQ_t = 0$.

Thus, the final expression is:

$$\nu_t^* = \frac{\sqrt{k\varphi}}{k} (1-\rho) \frac{\gamma e^{\xi(T-t)} + e^{-\xi(T-t)}}{\gamma e^{\xi(T-t)} - e^{-\xi(T-t)}} Q_t - \frac{\varphi}{k} \int_t^T \ell(u, t) ((1-\rho)Q_0 - \rho \mathbb{E}_{t,V}[V_u]) du$$

Recall $\frac{\sqrt{k\varphi}}{k} (1-\rho) = \xi$

$$\nu_t^* = \xi \frac{1+\gamma}{\gamma e^{\xi(T-t)} - e^{-\xi(T-t)}} Q_t - \frac{\xi^2}{1-\rho} \int_t^T \ell(u, t) ((1-\rho)Q_0 - \rho \mathbb{E}[V_u | \mathcal{F}_t^V]) du.$$

Intuition.

- The first term is a liquidation rate measuring time urgency: as $T-t \rightarrow 0$ the coefficient blows up, forcing any residual inventory to zero.
- The integral term anticipates future market volume. If $\mathbb{E}[V_u]$ is large, the bracket is expected to be large and trading rate increases to catch up with the liquidity. If volume is expected thin then it prompts the trader to trade at lower rate to wait for better liquidity.

(d) Implementation & Calibration with Real or Simulated Data

(i) Modelling and simulating cumulative market volume V_t

1. Given historical intraday volume data or simulated data, compute average cumulative volume curve across days. Usually will be U shape.
2. Use this U shape curve as a baseline profil for V_t .

3. Model difference between historical V_t and the fitted U curve by processes like GBM or OU process.
4. Compute sample variance of these differences to calibrate noise term σ .
5. Simulate paths by adding noise from 4. to deterministic baseline profile in 2. Can test optimal trading strategy in this environment.

(ii) Estimating model parameters

Parameter	Interpretation	Estimation / choice
σ	price volatility	realised std. of mid-price 1-min returns
b	permanent impact	regression $\Delta S \sim$ signed volume imbalance
k	temporary impact	slope of average slippage vs. execution size
φ	VWAP tracking penalty	grid-search on back-test (PNL vs. tracking error)
α	terminal inventory penalty	risk-tolerance setting; tune until $Q_T \approx 0$ in sims

Can backtest the model with difference choices of parameters to see which one yields robust trading strategies,

(iii) Inputs, statistical methods, simplifying assumptions

- **Inputs.** Level-1 trade/quote (TAQ) or full limit-order-book (LOB) updates; historical intra-day volume; mid-price series.
- **Methods.** OLS regression for b, k ; moving-average for smoothing μ_t ; cross-validation / grid-search for (φ, α) ; Monte-Carlo stress tests.
- **Simplifying assumptions.**
 - Immediate execution at effective price $\hat{S}_t = S_t - k\nu_t$ (ignore latency).
 - Parameters are stationary over calibration, can adjust and update periodically if needed.
 - Price impact is linear in trade size.
 - Ignore fixed transaction fees in first pass.

(e) Extra-credit implementation

Full numerical implementation in Python code of the VWAP model is provided separately as `vwap_simulation.py`, based on the planning in part (d).

(i) Simulated cumulative market volume V_t

Figure 1 shows a path for cumulative V_t and a path for deterministic U-shaped instantaneous V_t with added noise.

(ii) Model parameters used

The following values were selected for the structural and control parameters:

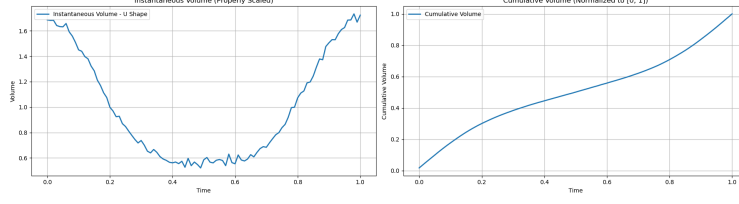


Figure 1: Simulated instantaneous and cumulative volume V_t

$$\begin{array}{ll}
 \sigma &= 0.0200 & \varphi &= 1.0000 \\
 b &= 0.0100 & \alpha &= 0.5000 \\
 k &= 0.1000 & \rho &= 0.5000 \\
 \xi &= 1.5811 & \gamma &= 1.9387
 \end{array}$$

(iii) Optimal trading speed ν_t^*

Figure 2 displays the optimal trading speed path computed using the closed-form expression $\nu^* = -\frac{bq+h_1+2h_2q}{2k}$ derived in previous part.

This incorporates both an inventory liquidation term and a forward-looking VWAP correction. The simulation code is attached as `vwap_simulation.py`.

(iv) Inventory path Q_t

Figure 2 shows:

- The evolution of inventory Q_t which monotonically decline to near zero as trading progresses.
- The evolution of optimal trading rate ν^*

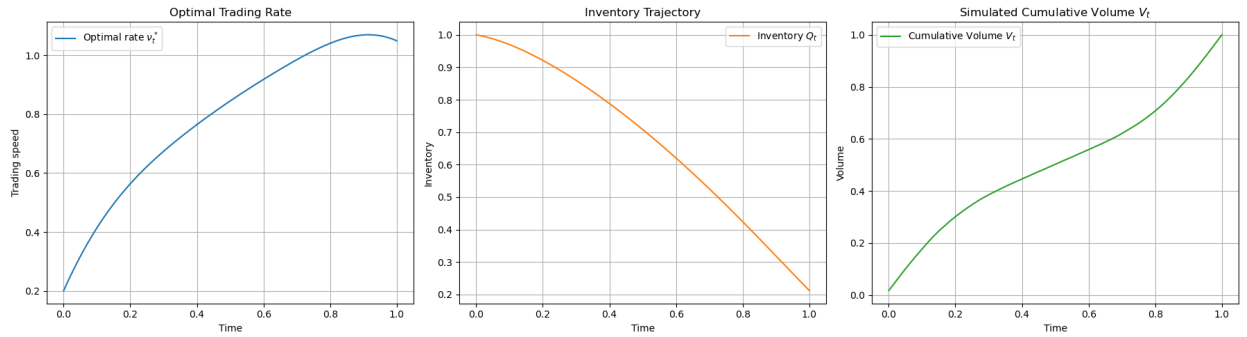


Figure 2: Optimal trading rate ν_t^* over time; Inventory trajectory Q_t over time

(v) Discussion of behavior

- The optimal trading rate ν^* peaks before the end but decreases slightly as trading approaches the final time as inventory depletes.
- Trading rate adapts to market volume. It increases during periods of higher market volume (e.g., time 0.0 - 0.2 and 0.8–1.0), as seen by the steeper slope in cumulative V_t . Specifically, ν^*

increases fast during period 0.0 -0.2 and peaks in 0.8-1.0. The strategy concentrates trading when the market can better absorb to minimize impact costs.

Problem 3

(a) How to use stochastic control strategy to design a trading trajectory under Guaranteed VWAP

1. Use

$$\nu_t^* = \xi \frac{1 + \gamma}{\gamma e^{\xi(T-t)} - e^{-\xi(T-t)}} Q_t - \frac{\xi^2}{1 - \rho} \int_t^T \ell(u, t) ((1 - \rho)Q_0 - \rho \mathbb{E}[V_u | \mathcal{F}_t^V]) du$$

as dynamic execution policy. Specifically, at each time t :

- Observe Q_t , current inventory
- observe V_t , realized cumulative volume
- update $\mathbb{E}[V_u | \mathcal{F}_t^V]$ of future volume for all $u \geq t$
- Compute ν^*

Ideally, one can trade infinitesimally at continuously-updated ν^* ,

$$Q_{t+dt} = Q_t - \nu_t^* dt.$$

2. Using client's VWAP as benchmark, then the random pL from the guarantee would be $\mathcal{P} = Q_0 \text{VWAP}_{[0,T]} - C(\nu^*)$. where C is the implementation shortfall.

3. Adapting cost for the Guaranteed VWAP contract is $E + LV$, we have stochastic control based guarantee cost as

$$\text{Guarantee Price} = \mathbb{E}[C(\nu) - Q_0 \text{VWAP}] + \lambda \text{Var}[C(\nu) - Q_0 \text{VWAP}]$$

, expected cost plus risk aversion times risk.

4. Can use Monte Carlo simulation to price Guarantee VWAP. Specifically:

- Generate many sample paths where each path consists of two time series on the interval $[0, T]$: V_t path and S_t path.
- On each path, one can apply trading strategy ν^* to compute implementation shortfall $\mathcal{C}$ and build up its distributions from Monte Carlo simulations.
- Estimate $\mathbb{E}[\mathcal{C}]$ and $\text{Var}[\mathcal{C}]$
- As mentioned before, guarantee cost is then $= \mathbb{E}[\mathcal{C}] + \lambda \text{Var}(\mathcal{C})$

(b) How can continuous time optimal control strategy improve trading approach compared to simulation-based approach in Question 1

For simplicity, continuous time optimal control strategy is referred as CTOC, and simulation-based strategy is SB.

1. SB uses brute force, while CTOC solution is closed form ν^* so parameter tuning is faster.
2. SB is in discrete time, while CTOC is continuous that makes trading rate more adaptive and further optimize trading costs.
3. SB needs to re-estimate future volume after each bucket, while CTOC employs Bayesian updating rule $\mathbb{E}[V_u | \mathcal{F}_t]$.

(c) Limitations and challenges of adapting stochastic control to discrete bucket-based simulation

1. Optimal trading rate ν^* changes continuously, but in a discrete bucket scheme, the rate is step-wise flat. This deviations introduce tracking error.
2. The control law's second term $\int_t^T \ell(u, t) \mathbb{E}[V_u | \mathcal{F}_t^V] du$ must be approximated by a Riemann sum over future buckets. Coarser buckets introduces larger quadrature error in estimating that integral.
3. Computing $\mathbb{E}[V_u | \mathcal{F}_t]$ relies on volume-model's continuous-time dynamics. In discrete buckets, it may need to switch to simple extrapolations per bucket, which can bias the control.